Figure 1: The latest entrant to the FOSS community [Source: *itsfoss.com*]

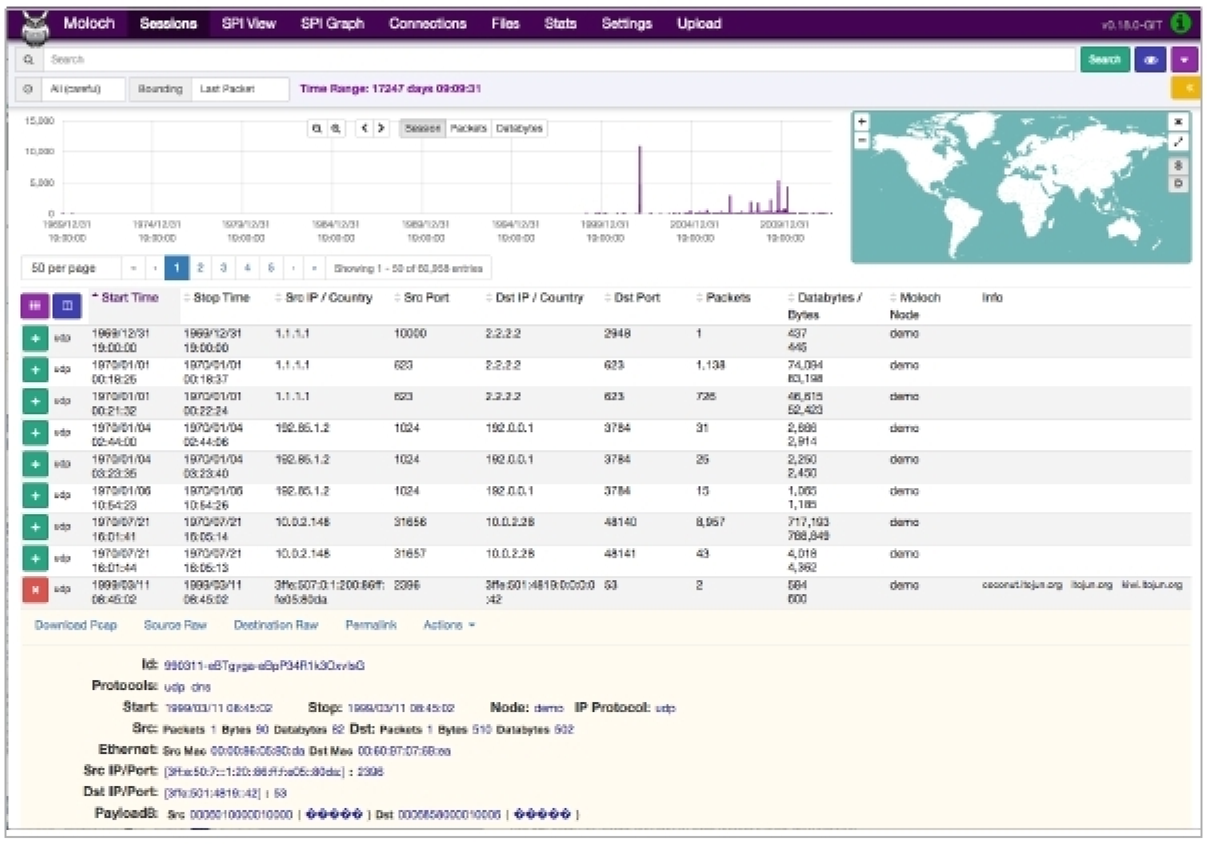

Figure 2: Moloch's visual interface [Source: *molo.ch*]

New age tools

Let's first look at the new entrants in the security domain, along with a detailed breakdown of their features and performance in creating a more secure network for enterprise and personal use cases.

Ghidra: The most recent contender on this list, NSA's Ghidra, is somewhat a surprise, last-minute entrant, considering it was only released a few days before the writing of this article. Not much is known about it yet, but at first glance, it seems to be a comprehensive reverse engineered framework for software, built to provide deep insight into malicious code as well as a better understanding of vulnerabilities in systems and computer networks.

To quote the GitHub description verbatim, this framework includes a suite of full-featured, high-end, software analysis tools that enable users to analyse compiled code on a variety of platforms including Windows, MacOS and Linux. Its capabilities include disassembly, assembly, decompilation, graphing and scripting, along with hundreds of other features. Ghidra supports a wide variety of process instruction sets and executable formats. It can be run in both user interactive and automated modes.

In addition, there is the functionality to allow users to develop their own Ghidra plugin in Java or Python. NSA says the entire idea behind its release was to foster an ecosystem and community efforts around this toolkit that would allow a better brand of talent to be available from the student and professional communities, leading to an overall benefit to the field. Sceptics claim this is a load of hogwash, but still don't know enough about Ghidra to counter the NSA's claims, effectively. The fact remains that the NSA has given us a lot to look into, and more opinions will most definitely emerge as the software tool is dissected and scrutinised further!

Osquery: Released by Facebook, Osquery uses SQL powered operating system instrumentation, monitoring, and analytics. Essentially, it exposes the operating system as a high performance relational database and allows you to write SQL based queries to explore the data within the operating system. In Osquery, the different processes, loaded kernel modules, open network connections, browser plugins and file hashes are treated as elements within an SQL table. It is fairly simple to then understand its schema and query the appropriate element or subset thereof.

Facebook claims there are two main features that users should know about Osquery:

- It runs everywhere and is entirely platform-independent due to there being no dependencies.
- It is fast and tested for memory leaks, thread safety and binary reproducibility on all supported platforms.

Moloch: AOL's Moloch is an open source, large scale, full packet-capturing, indexing and database system. The idea, as stated on its